
Cross-Species Transfer Learning with DeepSEA: From Human to Mouse

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Abstract

We test whether the convolutional features a DeepSEA model learns on the human genome transfer to mouse. We reuse the three convolutional layers of human DeepSEA, attach a fresh multi-task head, and fine-tune on roughly one million 1000-bp windows of mouse (mm10) ENCODE data spanning 133 chromatin features. Starting from human weights lowers validation loss faster and reaches a slightly higher held-out AUROC (0.877) than training the same network from scratch (0.865). The gap is small but consistent, which suggests that the early sequence-motif detectors learned on human are partly shared across species.

1 Introduction

DeepSEA is a convolutional neural network that predicts chromatin features such as transcription factor (TF) binding, DNase I hypersensitivity, and histone marks directly from DNA sequence [Zhou and Troyanskaya, 2015]. Its first convolutional layers act as motif detectors, learning short sequence patterns that resemble TF binding motifs. Because the regulatory code is built from such motifs, a natural question is whether these learned detectors are specific to the human genome or whether they capture patterns shared across species.

We study this through transfer learning from human to mouse. If the early layers of DeepSEA encode motifs that are conserved across species, then initializing a mouse model with human-trained convolutional weights should help, both by speeding up training and by improving accuracy, relative to training the same network from random initialization. A positive result would be evidence that DeepSEA learns common motifs rather than human-specific artifacts.

Related work. Our work builds directly on DeepSEA [Zhou and Troyanskaya, 2015], which introduced the architecture, the 1000-bp input window, and the multi-task formulation that predicts hundreds of chromatin features at once. DeepSEA was trained and evaluated only on the human genome. We keep its convolutional design and training setup and ask whether the features it learns extend to a second species.

2 Methods

Architecture. We split DeepSEA into a *trunk* and a *head*. The trunk is the three convolutional blocks of the original model: convolutions with 320, 480, and 960 kernels of width 8, each followed by a ReLU, with max-pooling (width 4) after the first two blocks and dropout throughout. The trunk maps a one-hot encoded 4×1000 input to a $960 \times L$ feature map. The head flattens this map and applies a linear layer with a sigmoid output per feature. The head is sized to the mouse feature set, so it is always trained from scratch. Only the trunk can carry transferred knowledge, because the human and mouse label sets do not line up one to one.

Transfer. Human weights come from the published DeepSEA checkpoint. The original weights are stored as 2D convolutions with a singleton dimension, so we squeeze them to 1D convolution

kernels and copy the three convolutional layers into our trunk by shape-matched order. We then compare two conditions that share all other settings: **Transfer**, where the trunk starts from human weights, and **Scratch**, where the trunk starts from random initialization. As a control we also report a **Baseline**, which uses the human trunk with an untrained head and is evaluated with no fine-tuning.

Data. We built a DeepSEA-style multi-task dataset for mouse (mm10) from ENCODE. We queried released narrowPeak files for TF ChIP-seq, histone ChIP-seq, and DNase-seq across five biosamples: liver, heart, forebrain, CH12.LX, and MEL. A feature is a (biosample, target) pair, for example `liver|CTCF`, so the same target in several cell types counts as several features. We binned the genome into 200-bp bins and labeled a bin positive for a feature when more than half of the bin overlaps a peak of that feature. Each retained bin was expanded to a 1000-bp window centered on it and one-hot encoded over $\{A, C, G, T\}$, with unknown bases mapped to an all-zero column. We subsampled to 10^6 windows over 133 features.

Training and evaluation. We split by chromosome to avoid leakage: chr19 is the test set and chr18 is the validation set, both held out entirely, leaving about 945,000 training windows. We train with AdamW and binary cross-entropy with logits, using a per-feature positive weight equal to the negative-to-positive ratio (clipped to $[1, 100]$) to handle class imbalance. We use a learning rate of 10^{-3} for the head, batch size 256, a plateau learning-rate scheduler, and early stopping on validation loss. We report per-feature AUROC and AUPRC on chr19, summarized by their median across the 132 features that have both positive and negative labels in the test chromosome. All models were trained on Google Colab using an NVIDIA L4 GPU.

3 Results

Transfer trains faster and scores slightly higher. Table 1 reports the held-out metrics for the three conditions. The transfer model reaches a median AUROC of 0.877, just above the from-scratch model at 0.865. The zero-shot baseline sits at chance (0.496), as expected, since its head is never trained. Figure 1 shows that the difference in training dynamics is clearer than the difference in final score: starting from human weights, the validation loss drops faster and reaches its minimum of 0.7277 at epoch 5, while the from-scratch model needs until epoch 12 to reach a slightly worse minimum of 0.7403. The human-initialized model also starts from a lower training loss in the first epoch, indicating that the transferred filters are already partly useful before any mouse fine-tuning.

Table 1: Held-out (chr19) performance, median over 132 evaluable features.

Condition	Trunk init	Fine-tuned	Median AUROC	Median AUPRC
Baseline	Human	No	0.496	0.018
Scratch	Random	Yes	0.865	0.230
Transfer	Human	Yes	0.877	0.250

Per-feature behavior matches the original DeepSEA. Figure 2 shows the distribution of per-feature AUROC for the transfer model. The model predicts TF and structural targets best: CTCF (across all cell types), RAD21, SMC3, USF1/2, POLR2A, and ZC3H11A all exceed an AUROC of 0.95. The hardest features are repressive and broad histone marks such as H3K27me3, H3K4me1, H3K36me3, and H3K79me2, which fall to 0.68 to 0.78. This ordering, sharp TF and accessibility signals easy and broad histone marks hard, is the same pattern reported for human DeepSEA [Zhou and Troyanskaya, 2015].

4 Discussion

Transfer learning from human to mouse works. Starting from human DeepSEA weights lowers loss faster and gives a more accurate model than random initialization. This supports the idea that the early convolutional layers learn motifs that are shared between the two species, rather than features that only apply to human sequence. We note that the accuracy gap is small, 0.877 versus 0.865 in median AUROC, so part of it could come from run-to-run randomness. The clearer and more reliable

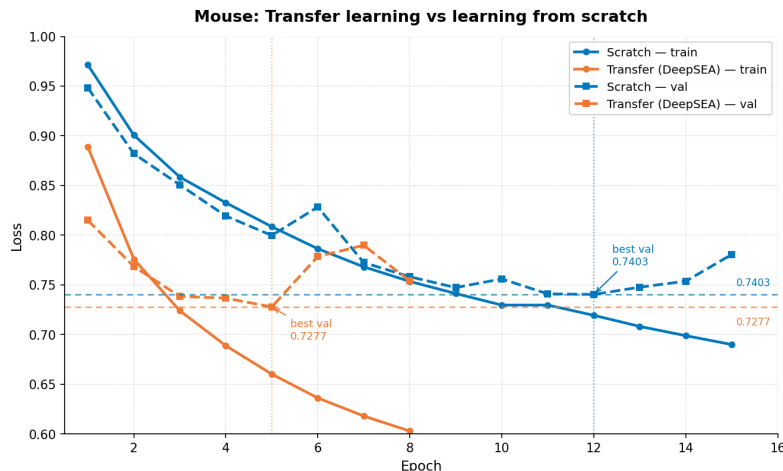


Figure 1: Training and validation loss for transfer versus from-scratch. The human-initialized model (orange) decreases loss faster and reaches a lower best validation loss (0.7277 at epoch 5) than the random-initialized model (blue, 0.7403 at epoch 12). Dashed horizontal lines mark each best validation loss.

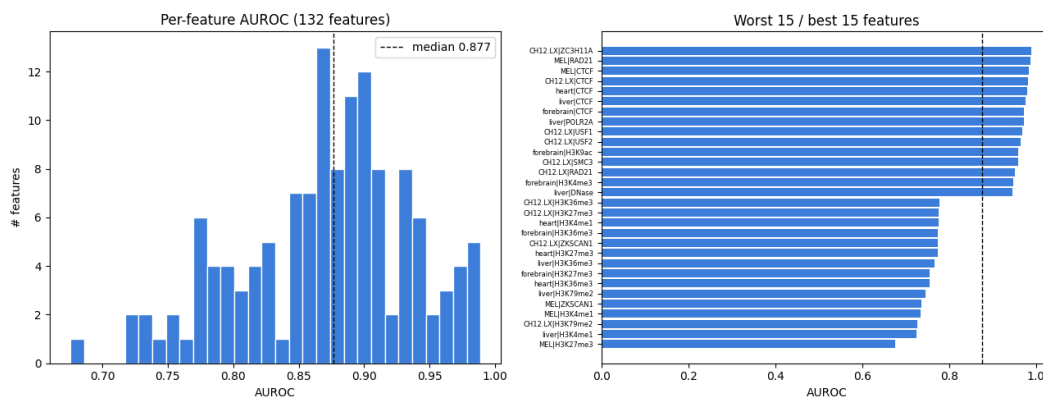


Figure 2: Per-feature AUROC of the transfer model on chr19. Left: distribution over 132 features, median 0.877. Right: the 15 best and 15 worst features. TF and structural targets (CTCF, RAD21, USF, POLR2A) score highest; broad histone marks score lowest.

effect is on optimization speed, where the human-initialized model reaches a better validation loss in fewer than half as many epochs.

Limitations. Our results are bounded by compute and by the amount of mouse data we trained on. We capped the dataset at 10^6 windows and stopped training early once validation loss plateaued, so we did not push either model to its best possible AUROC. Reaching higher accuracy would have required much longer training, which we could not afford on a single L4 GPU. This also limits how strongly we can claim the transfer gap is real, since a more fully trained pair of models might converge to similar accuracy even if transfer keeps its speed advantage. Our median AUROC of 0.877 is below the values reported for human DeepSEA, where median AUROC was about 0.958 for TF binding, 0.923 for DNase, and 0.856 for histone marks [Zhou and Troyanskaya, 2015]. The gap is consistent with our smaller training budget and smaller feature set rather than a difference in the method. We leave fuller training and the extension to more distant species for future work.

References

Jian Zhou and Olga G. Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12(10):931–934, 2015.